

# COMPLETED CHIRAL BAR-COBAR DUALITY AND MC<sub>4</sub> WEIGHT CUTOFF

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**ABSTRACT.** The classical bar-cobar adjunction of Priddy produces a Koszul dual for a quadratic algebra. For non-quadratic chiral algebras on a curve, completed bar constructions require two independent inputs: a degree cutoff for the Maurer-Cartan equation and a Mittag-Leffler condition for the inverse system of finite-stage cohomologies. A *strong completion tower* is a filtered augmented curved chiral  $A_\infty$ -algebra whose operations are filtration-nondecreasing on the augmentation ideal. Under this axiom the Maurer-Cartan equation is componentwise finite. If the finite-stage cohomology tower is Mittag-Leffler, the completed bar-cobar comparison is a quasi-isomorphism in the square-zero total model; in the genuinely curved CDG ambient it is a coacyclic-cone comparison in the second-kind coderived category.

The standard conformal filtrations on affine Kac-Moody, Virasoro, principal  $\mathcal{W}$ -algebras, and lattice vertex algebras do not by themselves supply the algebra-level strong filtration axiom. They enter through bar-level weight-cutoff criteria. The positive MC<sub>4</sub><sup>+</sup> regime is the  $\rho = 0$  case, proved by strong completion or by a genuine auxiliary positive grading. The generic finite-resonance MC<sub>4</sub><sup>0</sup> regime, including Virasoro and principal  $\mathcal{W}_N$  at generic parameters, is proved on the positive-energy weight-completed finite-resonance surface. A bare screening presentation still requires the explicit chain-level strong deformation retract or an independent stabilization theorem that identifies it with that surface.

## 1. INTRODUCTION

**1.1. The quadratic case and its limits.** Classical Koszul duality is a phenomenon of graded algebras. A quadratic algebra  $A = T(V)/(R)$  determines the dual coalgebra  $A^i \hookrightarrow T^c(sV^*)$ , the maximal subcoalgebra whose quadratic core is  $s^2 R^\perp$ ; the bar construction  $B(A)$  mediates between them; and  $A$  is Koszul when  $H^*(B(A))$  concentrates in bar length one, so that  $\Omega(A^i) \simeq A$ . The theory is due to Priddy [7], Beilinson–Ginzburg–Soergel [2], and in operadic form Loday-Vallette [4].

On an algebraic curve  $X$ , the objects of interest are chiral (vertex) algebras, whose relations are never quadratic in the strict sense. Heisenberg has a central term in the  $J \cdot J$  OPE. Virasoro has a quartic pole. Principal  $\mathcal{W}$ -algebras  $\mathcal{W}^k(\mathfrak{g}, f_{\text{prin}})$  have generators of weights  $m_i + 1$ , where  $m_i$  are the exponents of  $\mathfrak{g}$ ; only for  $\mathfrak{g} = \mathfrak{sl}_N$  are the weights  $2, \dots, N$ . Affine Yangians and quantum toroidal algebras have spectral-parameter relations of unbounded polynomial order. In each case the bar complex is a perfectly good dg coalgebra, but the iterated OPE produces bar elements whose differential contains infinitely many summands. A finite-dimensional truncation sees only a finite part of this object, and the classical bar-cobar adjunction is silent.

**1.2. The completion problem.** The remedy is completion. One introduces a decreasing filtration  $\mathcal{A} = F^0 \mathcal{A} \supset F^1 \mathcal{A} \supset \dots$  with finite-dimensional quotients  $\mathcal{A}_{\leq N} := \mathcal{A}/F^{N+1} \mathcal{A}$  and writes  $\widehat{B}(\mathcal{A}) := \varprojlim_N \widehat{B}(\mathcal{A}_{\leq N})$ . At each finite stage  $N$  the strict bar-cobar adjunction produces a universal twisting morphism  $\tau_N : \widehat{B}(\mathcal{A}_{\leq N}) \rightarrow \mathcal{A}_{\leq N}$  which is a Maurer-Cartan element of the convolution dg Lie algebra  $\text{Conv}(\widehat{B}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})$ . The completion problem, labelled MC<sub>4</sub> in [5], asks: do the finite-stage MC

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elements  $\tau_N$  glue to a single MC element  $\widehat{\tau}$  in the completed convolution algebra, and is the twisted tensor product acyclic in the limit?

Two obstructions appear. First, the MC equation  $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$  is a sum over all bar degrees, and a priori each component is infinite. Second, the passage from finite-stage acyclicity to acyclicity in the limit requires the Mittag-Leffler condition, which is a separate finite-window hypothesis.

**1.3. The main result.** The central structural axiom is that the  $A_\infty$  operations of  $\mathcal{A}$  respect the additive weight:

$$\mu_r(F^{i_1}\mathcal{A}, \dots, F^{i_r}\mathcal{A}) \subset F^{i_1+\dots+i_r}\mathcal{A}. \quad (1.1)$$

A filtered curved chiral  $A_\infty$ -algebra for which (1.1) holds and whose augmentation ideal  $\overline{\mathcal{A}} = F^1\mathcal{A}$  is the first filtration piece is a *strong completion tower* (Definition 3.1).

**Theorem 1.1** (Strong completion theorem, informal form). *Let  $\mathcal{A}$  be a strong completion tower whose finite-stage quotients lie in the proved square-zero bar-cobar regime and whose finite-stage cohomology tower is Mittag-Leffler. Then the completed bar construction  $\widehat{B}(\mathcal{A})$  is a separated complete pronilpotent curved dg chiral coalgebra, the completed convolution algebra carries a canonical Maurer-Cartan element  $\widehat{\tau} = \varprojlim_N \tau_N$ , and the counit*

$$\widehat{\epsilon}: \widehat{\Omega}(\widehat{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$$

*is a quasi-isomorphism in the square-zero total model. In the curved CDG ambient the same finite-stage construction gives a coacyclic-cone comparison under the second-kind hypotheses.*

The precise statement is Theorem 6.1 below. The proof uses three ingredients: (i) the degree cutoff lemma (§4), which reduces each component of the MC equation to a finite sum; (ii) the finite-window Mittag-Leffler criterion (§5); and (iii) the reduction principle of Lemma 5.2 which turns the finite-stage theory into a limit statement via the Milnor exact sequence.

**1.4. Positive versus resonant towers.** Completion problems split into positive and resonant regimes. A tower is *positive* if the filtration is defined by a genuine positive grading (e.g. the  $L_0$ -eigenvalue filtration on a graded vertex algebra). The degree cutoff lemma then holds after the strong axiom or the bar-level weight-cutoff criterion is verified. This is the  $\text{MC}_4^+$  regime.

A tower is *resonant* if there is a finite-dimensional weight-zero piece  $R_{\mathcal{A}}$  on which higher operations can preserve filtration degree. The *resonance rank*

$$\rho(\mathcal{A}) := \dim H^*(R_{\mathcal{A}}, m_1^{R_{\mathcal{A}}})$$

classifies the completion difficulty. Positive towers have  $\rho = 0$ ; Virasoro and the principal  $\mathcal{W}_N$ -algebras have  $0 < \rho < \infty$ ; genuinely wild algebras have  $\rho = \infty$ . The  $\text{MC}_4^0$  problem concerns the middle regime. For Virasoro and principal  $\mathcal{W}_N$  at generic parameters, the positive-energy weight-completed finite-resonance model proves the completed comparison. A concrete screening presentation of a principal  $\mathcal{W}_N$  algebra enters this theorem only after one gives the chain-level strong deformation retract, or an independent weight-cutoff stabilization theorem, identifying it with that finite-resonance model.

**1.5. Strong filtration in curved bar-cobar theory.** Positselski [6] developed curved bar constructions for associative algebras with curvature. Loday-Vallette [4] supply the operadic bar-cobar machinery on which the chiral lift is built. The Beilinson-Drinfeld framework of filtered chiral algebras with  $I$ -adic topology [1] provides the ambient topology for the inverse limit. The theorem below is the chiral and  $A_\infty$  lift: the novelty is the strong filtration axiom, which identifies the precise structural hypothesis under which all finite-stage data assemble without further input.

## 2. THE BAR CONSTRUCTION ON THE AUGMENTATION IDEAL

Throughout,  $X$  is a smooth complex algebraic curve and  $\mathcal{A}$  is an augmented curved chiral  $A_\infty$ -algebra on  $X$  in the sense of the monograph [5]. Augmentation means that  $\mathcal{A}$  is equipped with a counit  $\varepsilon: \mathcal{A} \rightarrow \mathbf{1}$  splitting the unit. The augmentation ideal is

$$\overline{\mathcal{A}} := \ker(\varepsilon).$$

*Remark 2.1* (Augmentation ideal). The bar construction is built on  $\overline{\mathcal{A}}$ , not on  $\mathcal{A}$ : writing  $T^c(s^{-1}\mathcal{A})$  in place of  $T^c(s^{-1}\overline{\mathcal{A}})$  includes the unit and breaks the Koszul sign computation. We therefore set

$$\overline{B}^{\text{ch}}(\mathcal{A}) := T^c(s^{-1}\overline{\mathcal{A}}),$$

with  $s^{-1}$  the homological desuspension  $|s^{-1}v| = |v| - 1$  and with the chiral coproduct obtained by deconcatenation on the collision divisor of  $\overline{C}_{\bullet+1}(X)$ .

The curved chiral  $A_\infty$  operations  $\{\mu_r\}_{r \geq 0}$  assemble into a codifferential  $d_{\overline{B}}$  on  $\overline{B}^{\text{ch}}(\mathcal{A})$  with curvature coming from  $\mu_0$ , so  $\overline{B}^{\text{ch}}(\mathcal{A})$  is a curved dg coalgebra. The convolution dg Lie algebra  $\text{Conv}(\mathcal{C}, \mathcal{A}) := \text{Hom}(\mathcal{C}, \mathcal{A})$  has bracket from the coproduct on  $\mathcal{C}$  and the product on  $\mathcal{A}$ ; a Maurer-Cartan element is a degree-1 map  $\tau: \mathcal{C} \rightarrow \mathcal{A}$  with  $\partial(\tau) + \tau \star \tau = 0$ .

**Theorem 2.2** (Finite-type bar-cobar duality, cited). *Let  $\mathcal{A}$  be a finite-type augmented curved chiral  $A_\infty$ -algebra on  $X$  in the proved square-zero bar-cobar regime. Then the universal twisting morphism  $\tau: \overline{B}^{\text{ch}}(\mathcal{A}) \rightarrow \mathcal{A}$  is a Maurer-Cartan element of  $\text{Conv}(\overline{B}^{\text{ch}}(\mathcal{A}), \mathcal{A})$ , and the counit*

$$\epsilon: \Omega^{\text{ch}}(\overline{B}^{\text{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$$

*is a quasi-isomorphism in the square-zero total model. In the curved CDG ambient the corresponding finite-stage cone is coacyclic in the second-kind coderived category.*

*Proof.* See [5, Thm. B]. The finite-type hypothesis reduces to finite-type vertex algebras, where the operadic bar-cobar machinery of Loday-Vallette [4] applies after the curved extension of Positselski [6].  $\square$

## 3. THE STRONG FILTRATION AXIOM

**Definition 3.1** (Strong completion tower). An augmented curved chiral  $A_\infty$ -algebra  $\mathcal{A}$  on a curve  $X$  is a *strong completion tower* if it carries a descending filtration

$$\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset F^2\mathcal{A} \supset \dots, \quad \bigcap_{N \geq 0} F^{N+1}\mathcal{A} = 0,$$

such that:

- (i)  $\mathcal{A}$  is separated and complete:  $\mathcal{A} \cong \varprojlim_N \mathcal{A}_{\leq N}$ , where  $\mathcal{A}_{\leq N} := \mathcal{A}/F^{N+1}\mathcal{A}$ ;
- (ii) every quotient  $\mathcal{A}_{\leq N}$  is of finite type and lies in the proved bar-cobar regime of Theorem 2.2;
- (iii)  $\overline{\mathcal{A}} = F^1\mathcal{A}$ : the augmentation ideal is the first filtration piece;
- (iv) the curvature term satisfies  $\mu_0(1) \in F^1\mathcal{A}$ , and all positive-arity chiral  $A_\infty$ -operations are filtration-nondecreasing,

$$\mu_r(F^{i_1}\mathcal{A}, \dots, F^{i_r}\mathcal{A}) \subset F^{i_1 + \dots + i_r}\mathcal{A} \tag{3.1}$$

for  $r \geq 1$ .

Axiom (3) ties the filtration to the augmentation. Axiom (4) is the structural heart of the definition: the curvature starts in the augmentation ideal, and the positive-arity operations respect the additive grading.

*Remark 3.2* (On the quotient maps). Each projection  $p_N: \mathcal{A}_{\leq N+1} \rightarrow \mathcal{A}_{\leq N}$  is a strict morphism of curved chiral  $A_\infty$ -algebras, because the kernel  $F^{N+1}\mathcal{A}/F^{N+2}\mathcal{A}$  is an  $A_\infty$ -ideal of  $\mathcal{A}_{\leq N+1}$  by (3.1). The finite-stage bar constructions  $\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N})$  therefore form an inverse system of curved dg chiral coalgebras with strict morphisms, and the completed bar construction is the literal inverse limit.

**Proposition 3.3** (Standard landscape weight cutoff). *For the standard conformal-weight filtration  $F^N \mathcal{A} := \bigoplus_{h \geq N} \mathcal{A}_h$ , the following families satisfy the bar-level weight-cutoff criterion: in every fixed total conformal weight only finitely many generators and OPE residue terms contribute to the completed bar differential.*

- (a) the affine Kac-Moody vertex algebra  $V_k(\mathfrak{g})$  for any simple  $\mathfrak{g}$  and any non-critical level  $k \neq -h^\vee$ ;
- (b) the Virasoro vertex algebra  $\text{Vir}_c$  for any  $c$ ;
- (c) the principal  $\mathcal{W}$ -algebra  $\mathcal{W}^k(\mathfrak{g}, f_{\text{prin}})$  for any simple  $\mathfrak{g}$  and any non-critical  $k$ ;
- (d) the lattice vertex algebra  $V_\Lambda$  for any positive-definite even lattice  $\Lambda$ .

*This statement does not assert the algebra-level strong filtration axiom  $\mu_r(F^{i_1}, \dots, F^{i_r}) \subset F^{i_1 + \dots + i_r}$ .*

*Proof.* Each family is  $L_0$ -graded with finite-dimensional homogeneous pieces and finitely many strong generators in each bounded weight window. The OPE of two homogeneous fields of weights  $a, b$  has only finitely many poles, and every residue term has conformal weight at most  $a + b - 1$ . Thus a bar chain of fixed total weight uses only fields of bounded weight and only finitely many residue terms. This is the bar-level cutoff needed for the finite-window MC4 comparison.

The assertion is weaker than algebra-level strong filtration. For affine currents, for example, the term  $[J^a, J^b]$  has weight 1 although the inputs have weights 1 and 1; hence the strong axiom fails for the raw conformal filtration. Principal  $\mathcal{W}^k(\mathfrak{g}, f_{\text{prin}})$  is governed by the exponents of  $\mathfrak{g}$ : the generator weights are  $m_i + 1$ , with  $m_i$  running through the exponents.  $\square$

**Example 3.4** (Non-example:  $\beta\gamma$ -system with conformal grading). The  $\beta\gamma$ -system has a weight- $\frac{1}{2}$  grading that refines the conformal grading. With respect to the conformal-weight filtration alone, the strong filtration axiom fails because the OPE  $\beta(z)\gamma(w) \sim 1/(z-w)$  produces a weight-zero output from weight- $\frac{1}{2}$  inputs, violating additivity. Its shadow tower is contact class  $C$ , with a nonzero quartic contact shadow; it is not a Heisenberg-type degree-2 tower. A different auxiliary filtration may still give a bar-level weight cutoff.

#### 4. THE DEGREE CUTOFF LEMMA

The structural consequence of the strong filtration axiom is that in the finite-stage bar complex, only finitely many degrees contribute. This is the precise mechanism by which the MC equation becomes componentwise finite.

**Lemma 4.1** (Degree cutoff). *Let  $\mathcal{A}$  be a strong completion tower and let  $\mathcal{A}_{\leq N} = \mathcal{A}/F^{N+1}\mathcal{A}$  be the  $N$ -th quotient. In the finite-stage convolution algebra  $\text{Conv}(\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})$ , the Maurer-Cartan equation*

$$\partial(\tau_N) + \tau_N \star \tau_N = 0$$

*involves only degrees  $r \leq N$ : on  $\mathcal{A}_{\leq N}$  the bar differential is the finite sum  $b_0 + b_1 + \dots + b_N$  of coderivations coming from  $\mu_0, \mu_1, \dots, \mu_N$ .*

*Proof.* The arity-zero curvature term  $b_0$  is a single continuous term because  $\mu_0(1) \in F^1\mathcal{A}$ . Every positive-arity bar input lies in  $\overline{\mathcal{A}}_{\leq N}$ , which by axiom (3) equals the image of  $F^1\mathcal{A}$  in the quotient. The strong filtration axiom (3.1) gives

$$\mu_r(\overline{\mathcal{A}}^{\otimes r}) \subset \mu_r((F^1\mathcal{A})^{\otimes r}) \subset F^r\mathcal{A}.$$

Reducing modulo  $F^{N+1}\mathcal{A}$ , any term with  $r \geq N+1$  lands in  $F^r\mathcal{A} \subset F^{N+1}\mathcal{A}$  and vanishes in  $\mathcal{A}_{\leq N}$ . So the induced bar differential on  $\mathcal{A}_{\leq N}$  is the finite sum  $b_0 + b_1 + \cdots + b_N$ , as claimed.  $\square$

*Remark 4.2* (Degree cutoff as PBW degeneration). The degree cutoff is the operadic counterpart of the classical PBW theorem. For the universal enveloping algebra  $U(\mathfrak{g})$  with its standard filtration by number of generators, the associated graded is  $\text{Sym}(\mathfrak{g})$ ; higher-degree brackets vanish modulo lower filtration stages. The strong filtration axiom extends this phenomenon to all chiral  $A_\infty$ -operations on a graded vertex algebra: the degree cutoff lemma is the statement that at each finite stage, only finitely many Poisson (or vertex) operations are active. The classical PBW theorem is the  $r = 2$  case.

**Corollary 4.3** (Finite-stage MC assembly). *Let  $\mathcal{A}$  be a strong completion tower. The universal twisting morphism  $\tau_N$  at stage  $N$  is a Maurer-Cartan element of the finite-stage, locally finite convolution  $A_\infty$ -algebra; the quotient  $\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N})$  has finite-dimensional bar-length components, and the bar differential is the finite sum of Lemma 4.1.*

*Proof.* Finite type of  $\mathcal{A}_{\leq N}$  gives finite dimension of each  $\overline{\mathcal{A}}_{\leq N}^{\otimes r}$  for every  $r$ , and the degree cutoff lemma bounds the number of nonzero bar differential terms. Theorem 2.2 then gives the Maurer-Cartan property.  $\square$

## 5. MITTAG-LEFFLER FINITE-WINDOW CRITERION

The second obstruction to passing from finite-stage acyclicity to acyclicity in the limit is the possible failure of the Mittag-Leffler condition on the inverse system of finite-stage cohomologies. The strong filtration axiom supplies the degree cutoff; the Mittag-Leffler condition is a separate finite-window hypothesis.

**Proposition 5.1** (Mittag-Leffler criterion for finite windows). *Let  $\mathcal{A}$  be a filtered tower whose finite-stage bar complexes have finite weight windows: for each total weight  $w$  there is  $N(w)$  such that the weight- $w$  subcomplex of  $\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N})$  is independent of  $N$  for  $N \geq N(w)$ . Then, for every cohomological degree  $m$ , the inverse system*

$$\{H^m(\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N}))\}_{N \geq 0}$$

*is Mittag-Leffler on every fixed weight slice. Consequently the finite weight truncations, and the product completion over weight slices, have vanishing  $\varprojlim^1$ .*

*Proof.* The transition maps are surjections because the finite-stage bar construction is functorial on surjections of finite-type curved chiral  $A_\infty$ -algebras. By hypothesis, for each fixed weight  $w$  there is  $N(w)$  such that the transition map

$$\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N+1})_w \xrightarrow{\sim} \overline{B}^{\text{ch}}(\mathcal{A}_{\leq N})_w$$

is an isomorphism of subcomplexes for  $N \geq N(w)$ . Passing to cohomology gives stable isomorphisms on the weight- $w$  slice.

For a finite weight truncation only finitely many slices occur, so the inverse system is Mittag-Leffler. For the completed product over all weights, inverse limits and products are taken slice-wise over vector spaces; the preceding stabilization kills the derived limit on each slice, hence on the completed product.  $\square$

**Lemma 5.2** (Complete filtered comparison). *Let  $\{f_N: C_N \rightarrow D_N\}$  be a morphism of inverse systems of complexes with surjective transition maps. If each  $f_N$  is a quasi-isomorphism and the inverse systems  $\{H^*(C_N)\}$  and  $\{H^*(D_N)\}$  are Mittag-Leffler, then the induced map on limits  $\widehat{f}: \widehat{C} \rightarrow \widehat{D}$  is a quasi-isomorphism.*

*Proof.* For any inverse system of cochain complexes the Milnor exact sequence reads

$$0 \rightarrow \varprojlim_N^1 H^{m-1}(C_N) \rightarrow H^m(\widehat{C}) \rightarrow \varprojlim_N H^m(C_N) \rightarrow 0.$$

Mittag-Leffler kills the derived limit term, and the map on honest limits is an isomorphism because each  $f_N$  is.  $\square$

## 6. THE MAIN THEOREM

**Theorem 6.1** (Strong completion tower bar-cobar duality). *Let  $\mathcal{A}$  be a strong completion tower (Definition 3.1) whose finite-stage comparison systems satisfy the finite-window Mittag-Leffler condition of Proposition 5.1. Write  $\tau_N \in \text{MC}(\text{Conv}(\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N}))$  for the universal twisting morphism at stage  $N$ . Then:*

- (i) Completed coalgebra. *The completed bar construction*

$$\widehat{\overline{B}}^{\text{ch}}(\mathcal{A}) := \varprojlim_N \overline{B}^{\text{ch}}(\mathcal{A}_{\leq N})$$

*exists as a separated complete pronilpotent curved dg chiral coalgebra with continuous differential, coproduct, and curvature. At each stage,  $\widehat{\overline{B}}^{\text{ch}}(\mathcal{A})/F^{N+1} \cong \overline{B}^{\text{ch}}(\mathcal{A}_{\leq N})$ .*

- (ii) Completed MC element. *The componentwise limit  $\widehat{\tau} = \varprojlim_N \tau_N$  is a Maurer-Cartan element of the completed convolution algebra*

$$\widehat{\text{Conv}} := \varprojlim_N \text{Conv}(\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N}),$$

*and the MC equation  $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$  converges in the inverse-limit topology, each component being a finite sum by the degree cutoff lemma.*

- (iii) Completed cobar. *The completed cobar object*

$$\widehat{\Omega}^{\text{ch}}(\widehat{\overline{B}}^{\text{ch}}(\mathcal{A})) := \varprojlim_N \Omega^{\text{ch}}(\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N}))$$

*is a separated complete curved dg chiral algebra, equal to the twisted tensor product  $\mathcal{A} \otimes_{\widehat{\tau}} \widehat{\overline{B}}^{\text{ch}}(\mathcal{A})$ .*

- (iv) Counit in the square-zero total model. *The completed counit*

$$\widehat{\epsilon}: \widehat{\Omega}^{\text{ch}}(\widehat{\overline{B}}^{\text{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$$

*is a quasi-isomorphism when both sides carry the square-zero total differential. In the genuinely curved CDG ambient, the same comparison is replaced by a coacyclic-cone statement in the specified second-kind coderived category.*

- (v) Dual comparison. *If  $C = \varprojlim_N C_{\leq N}$  is a separated complete pronilpotent curved dg chiral coalgebra whose finite quotients lie in the finite-type square-zero bar-cobar regime and whose finite-window tower is Mittag-Leffler, then the completed unit*

$$\widehat{\eta}: C \xrightarrow{\sim} \widehat{\overline{B}}^{\text{ch}}(\widehat{\Omega}^{\text{ch}}(C))$$

*is a quasi-isomorphism in the square-zero total model, with the curved CDG version read as a coacyclic-cone comparison.*

- (vi) Equivalence surface. *Completed bar and completed cobar give quasi-inverse equivalences on the homotopy category of strong completion towers satisfying the same square-zero and Mittag-Leffler hypotheses. In a curved CDG realization the equivalence is the corresponding second-kind coderived equivalence.*

*Proof. Completed coalgebra.* Each quotient map  $p_N: \mathcal{A}_{\leq N+1} \rightarrow \mathcal{A}_{\leq N}$  is strict by axiom (4), inducing a strict morphism  $q_N: \overline{B}^{\text{ch}}(\mathcal{A}_{\leq N+1}) \rightarrow \overline{B}^{\text{ch}}(\mathcal{A}_{\leq N})$  compatible with differential, coproduct, coaugmentation, and curvature. The inverse limit exists componentwise with  $b = \varprojlim b_N$ ,  $\Delta = \varprojlim \Delta_N$ ,  $h = \varprojlim h_N$ ; the curved coalgebra identities hold quotientwise and hence on the limit. Continuity is formal for the inverse-limit topology. Pronilpotence: the reduced coproduct on bar-length  $\leq m$  elements is nilpotent on each finite quotient, so the limit is topologically pronilpotent.

*MC assembly.* Define  $\widehat{\tau} := \varprojlim_N \tau_N$ . By Lemma 4.1, modulo  $F^{N+1}\mathcal{A}$  only degrees  $\leq N$  contribute to  $\widehat{\tau} \star \widehat{\tau}$ . The MC equation holds modulo  $F^{N+1}$  (where it reduces to the finite-stage equation for  $\tau_N$ ), hence on the completed coalgebra. This is claim (2).

*Completed cobar.* Each  $\Omega^{\text{ch}}(\overline{B}^{\text{ch}}(\mathcal{A}_{\leq N}))$  is a curved dg chiral algebra; compatible transition maps give a complete curved dg chiral algebra in the limit, with continuous operations inherited from the finite stages. This is claim (3).

*Counit.* The quotient of  $\widehat{\epsilon}$  modulo  $F^{N+1}$  is the finite-stage counit  $\epsilon_N$ , which is a quasi-isomorphism by Theorem 2.2. The finite-window hypothesis gives the Mittag-Leffler condition for the source and target cohomology systems in Lemma 5.2. Applying that lemma to the compatible morphisms  $\{\epsilon_N\}$  yields claim (4).

*Dual comparison.* The argument for  $\widehat{\eta}$  is identical with bar and cobar exchanged, giving claim (5).

*Equivalence.* Claims (4) and (5) are the two triangle identities up to quasi-isomorphism, so completed bar and completed cobar form an adjoint equivalence on the homotopy categories restricted to strong completion towers. This is claim (6).  $\square$

## 7. POSITIVE AND RESONANT COMPLETION

The strong-completion theorem proves the formal MC<sub>4</sub> mechanism once the degree cutoff, square-zero or coderived ambient, and Mittag-Leffler hypotheses are in place. The positive regime is the case  $\rho(\mathcal{A}) = 0$ : every completed operation either strictly raises the chosen positive filtration, or an auxiliary positive grading gives the same finite-window statement.

**Corollary 7.1** (Positive MC<sub>4</sub>). *Let  $\mathcal{A}$  be a positive strong completion tower with  $\rho(\mathcal{A}) = 0$ , or a tower equipped with a genuine auxiliary positive grading whose bar differential satisfies the finite-window cutoff and the square-zero or coderived ambient hypotheses of Theorem 6.1. Then the completed bar-cobar counit is a quasi-isomorphism in the square-zero total model, and a coacyclic-cone comparison in the curved CDG ambient.*

*Proof.* The strong case is Theorem 6.1. In the auxiliary positive grading case, positivity supplies the finite-window hypothesis of Proposition 5.1, so the same inverse-limit comparison applies. Finite-resonance towers with  $0 < \rho(\mathcal{A}) < \infty$  are not included in this corollary.  $\square$

**Definition 7.2** (Finite resonance). A filtered chiral algebra has finite resonance if the part of the bar differential preserving filtration degree is controlled by a finite-dimensional complex  $R_{\mathcal{A}}$ . Its resonance rank is

$$\rho(\mathcal{A}) = \dim H^*(R_{\mathcal{A}}).$$

The equality  $\rho = 0$  is the positive case. The case  $0 < \rho < \infty$  is a finite-dimensional obstruction problem requiring explicit homotopy data.

**Theorem 7.3** (Generic finite-resonance  $\text{MC}_4^\circ$ ). *Let  $\mathcal{A}$  be Virasoro at generic central charge or a principal  $\mathcal{W}_N$  algebra at generic shifted level, equipped with its positive-energy weight-completed finite-resonance model. Assume the resonance complex  $R_{\mathcal{A}}$  is finite dimensional and is represented inside the completed bar complex by either a chain-level strong deformation retract or a finite-window stabilization. Then the  $\text{MC}_4^\circ$  completed bar-cobar comparison holds in the same square-zero, respectively coderived, ambient as Theorem 6.1.*

*Proof.* The positive-energy filtration decomposes the completed bar complex into finite weight windows. The part of the differential preserving filtration is, by hypothesis, represented by the finite complex  $R_{\mathcal{A}}$ ; the remaining summands strictly raise weight and therefore satisfy the cutoff of Proposition 5.1. The chosen strong deformation retract, or equivalently the finite-window stabilization, identifies the finite-resonance summand with a fixed finite complex at every sufficiently large stage. Hence the inverse system of finite-stage cohomologies is Mittag-Leffler: the positive part stabilizes weight by weight, and the resonance part is finite dimensional. The completed comparison is then the inverse-limit comparison of Theorem 6.1. The coderived statement is obtained by replacing the square-zero quasi-isomorphism with the same finite-stage coacyclic-cone comparison before taking the Mittag-Leffler limit.  $\square$

**Proposition 7.4** (Screening obstruction for the principal  $\mathcal{W}_N$  route). *Let  $\mathfrak{h}$  be the rank- $(N-1)$  Heisenberg vertex algebra with primitive chiral coproduct  $\Delta_z^{\mathfrak{h}}$ , and let  $Q_{\alpha_i}$  be the Feigin–Frenkel screening charge [3] at shifted level  $\Psi$ . A strong deformation retract presenting*

$$\mathcal{W}_N = \bigcap_i \ker Q_{\alpha_i} \subset \mathfrak{h}$$

*must account for the chiral 1-cocycle*

$$[Q_{\alpha_i}, \Delta_z^{\mathfrak{h}}] = \frac{\Psi-1}{\Psi} (: e^{\alpha_i \cdot \varphi(0)} :) \otimes (: e^{\alpha_i \cdot \varphi(-z)} :) \cdot (1 + O(z)).$$

*At  $\Psi = 1$  this obstruction vanishes. At generic  $\Psi$  it is nonzero and must vanish under the homotopy data or be absorbed in a completed coderived ambient.*

*Proof.* This is the Vol. I screening-coproduct computation: expanding the primitive Heisenberg coproduct on the mode decomposition of  $: e^{\alpha_i \cdot \varphi(\zeta)} :$  gives the one-loop contraction  $\alpha_i^2 \log z = (2/\Psi) \log z$ , whose regularized residue is  $(\Psi-1)/\Psi$ . The same coefficient is the Miura cross-term coefficient for the spin- $s$  principal  $\mathcal{W}_N$  generators.  $\square$

## 8. STANDARD FAMILIES

**8.1.  $\beta\gamma$  and affine Kac-Moody.** For the  $\beta\gamma_\lambda$  system the conformal weights are  $\lambda$  and  $1-\lambda$ ; the physical  $\beta\gamma$  point  $\lambda = 1$  gives weights 1 and 0. The raw conformal filtration therefore is not a positive strong completion tower. Its shadow data are contact class  $C$ :

$$S_3 = 0, \quad S_4 = -\frac{5}{12}, \quad S_r = 0 \quad (r \geq 5).$$

Any  $\text{MC}_4$  application must use an auxiliary filtration that proves the relevant bar-level cutoff.

For  $V_k(\mathfrak{g})$  at non-critical level the conformal filtration gives the finite-window bar cutoff of Proposition 3.3. The algebra-level strong filtration fails because the  $J_{(0)}^a J^b$  bracket has weight 1 from two weight-1 inputs. The critical level  $k = -h^\vee$  lies outside the finite-stage proved regime because the Feigin–Frenkel centre changes the completion problem.

**8.2. Virasoro and principal  $\mathcal{W}$ .** Virasoro has one weight-2 generator and OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$



The central fourth-order pole is the finite resonance direction. Generic Virasoro MC<sub>4</sub><sup>o</sup> is therefore the finite-resonance case of Theorem 7.3.

For principal  $\mathcal{W}^k(\mathfrak{g}, f_{\text{prin}})$  the generator weights are the exponents  $m_i + 1$ . In type  $A_{N-1}$  these are  $2, \dots, N$ . The standard principal-stage  $\mathcal{W}_\infty$  tower has a proved bar-level weight-cutoff route when the quotient system and square-zero/coderived hypotheses are specified. A generic Feigin–Frenkel screening descent for individual principal  $\mathcal{W}_N$  stages must still confront Proposition 7.4.

| Algebra                                          | Input                    | Conclusion             | Regime                       |
|--------------------------------------------------|--------------------------|------------------------|------------------------------|
| $V_k(\mathfrak{g}), k \neq -h^\vee$              | bar-level cutoff         | proved with hypotheses | MC <sub>4</sub> <sup>+</sup> |
| $V_\Lambda$                                      | bar-level cutoff         | proved with hypotheses | MC <sub>4</sub> <sup>+</sup> |
| $\beta\gamma_\lambda$                            | auxiliary contact cutoff | requires cutoff        | class $C$                    |
| Vir <sub>c</sub> generic                         | finite resonance model   | proved with hypotheses | MC <sub>4</sub> <sup>o</sup> |
| principal $\mathcal{W}_N$ generic                | finite resonance model   | proved with hypotheses | MC <sub>4</sub> <sup>o</sup> |
| principal $\mathcal{W}_N$ screening presentation | screening SDR            | requires SDR           | descent                      |
| $V_{-h^\vee}(\mathfrak{g})$                      | Feigin–Frenkel centre    | excluded               | critical                     |

## REFERENCES

- [1] A. Beilinson and V. Drinfeld, *Chiral algebras*, American Mathematical Society Colloquium Publications, vol. 51, American Mathematical Society, Providence, RI, 2004.
- [2] A. Beilinson, V. Ginzburg, and W. Soergel, *Koszul duality patterns in representation theory*, J. Amer. Math. Soc. **9** (1996), 473–527.
- [3] B. Feigin and E. Frenkel, *Affine Kac–Moody algebras at the critical level and Gelfand–Dikii algebras*, Internat. J. Modern Phys. A **7** (1992), suppl. 1A, 197–215.
- [4] J.-L. Loday and B. Vallette, *Algebraic operads*, Grundlehren der mathematischen Wissenschaften, vol. 346, Springer, Heidelberg, 2012.
- [5] R. Lorgat, *Modular Koszul Duality, Volume I*, Monograph, 2026.
- [6] L. Positselski, *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, Mem. Amer. Math. Soc. **212** (2011), no. 996.
- [7] S. B. Priddy, *Koszul resolutions*, Trans. Amer. Math. Soc. **152** (1970), 39–60.

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